

7 Cryptography

7.3 Public Key Cryptography

7.3.1 Key Exchange

Up to this point every method of sending an encrypted message requires a secret key known to both Alice and Bob. A big problem in cryptography is the following:
Suppose that Alice and Bob are unable to meet in person, they have to email each other to agree on the secret key. Unfortunately, Eve has access to Alice's emails, and thus can read any emails that are sent between Alice and Bob. How can Alice and Bob decide on a secret key without the key falling into Eve's hands?

We need some 'one-way function' that is easy to compute, but hard to undo. Illustrate how this can be done with the paint mixing example. In this example, the secret key is the final paint color.

Of course, since Alice and Bob are using computers, the one-way function will need to be a numerical function.

7.3.2 Modular Arithmetic

What is modular arithmetic?

Do clock example. $8 + 5 = 13$, we will write this as $8 + 5 \equiv 1 \pmod{12}$.

Definition 7.3.1 The **modulus** is another name for the remainder after division.

Example 7.3.2 Compute

1. $15 \bmod 7$
2. $125 \bmod 43$
3. $5 \bmod 20$
4. $-12 \bmod 7$
5. $2^7 \bmod 5$

Note that we will always write our final answer as a natural number between 0 and $n - 1$.

Example 7.3.3 Your turn! Compute the following

1. $123 \bmod 19$
2. $65 \bmod 12$
3. $-923 \bmod 134$
4. $-124 \bmod 543$
5. $1045 \bmod 521$
6. $8324 \bmod 358$